Question 1

You are a data consultant working for Go for Green Ltd (Green), an energy company based in Bristol, England. It currently has two million customers, who are predominantly based in the south of England. The company sells to customers directly through its website, or through comparison websites.

Green provides its customers solely with renewable energy. Currently, Green provides 100% renewable electricity from solar, wind, hydro and tidal and 100% of gas energy provided is carbon neutral. Carbon neutral means it offsets emissions from gas by supporting carbon-reducing projects around the world. Green publishes an annual stakeholder report which serves as a communication tool between the company and stakeholders. It contains information on Green's strategic objectives and provides stakeholders with several key performance indicators (KPIs) to provide transparency in how the company is performing.

Some examples are:

Objective	Ensure an even spread of energy sources to reduce risk
KPI	% Split of electricity energy provided by renewable category (solar, wind, hydro, tidal)

Objective	Increase households with smart meters
KPI	Number of customers with smart meters

Objective	Increase customer sales through Green's website
KPI	New customers per source (website directly, or comparison site)

Objective	Improve employee satisfaction
KPI	Existing employee satisfaction on a scale of 1 to 5

The KPI section of the annual report is created by Tina Strause, the Insight and Impact Manager. She has mentioned a couple of incidents recently and, as a result, is concerned about the integrity of the data being used to prepare the information. Tina has asked you to review some background information regarding Green's data processes (Appendix) and analyse any risks to the data integrity faced by Green, evaluating the impact on Green's ability to make effective strategic decisions.

Question:

Analyse the risks to data integrity faced by Green; and evaluate the impact of the risks on management's ability to design and implement an effective strategy.

(23 marks)

APPENDIX – Background information on Green's data processes Servers

Green has a physical server in the office in Bristol but has recently started moving data on to a cloud-based server. This was in anticipation of the Bristol server reaching capacity over the next year. Tina is going to use the data from the cloud-based server for her KPIs. Once all the data has been transferred across, the Bristol server will be decommissioned and no longer be in use. The transfer of data is taking place in stages. When customer application data was being transferred there was a power cut for 5 minutes in the office and some data did not transfer. It has not yet been established which data is not on the cloud-based server.

Data team

All Green's data is maintained by the data team, consisting of 10 staff. The department is headed up by two data managers and eight data assistants. All of the team have access to the servers using passwords, although certain sensitive data is accessible only by the data managers. Included in sensitive data is customer smart meter data, which has only recently been getting received now smart meters are being rolled out across their customers. Recently, two of the data assistants have been off sick and to help with the large workload, a work experience student was asked to help in the department. He was given full data manager access to the data.

Internal audit

Green has a well-established internal audit department which was set up by the board to gain assurance that the management and control of Green is robust. The data team have regularly engaged with internal audit and had various processes audited over the years without any issues being noted. Unusually, they have not had any internal audit reviews carried out in the last 18 months due to a company restructure and the internal audit department concentrating on other areas of the business.

Purchase system

Green partners with UK generators to directly purchase its electricity and gas. The transaction is recorded in a bespoke energy purchase system, which fully integrates with the accounting software. The purchase system contains the standard purchase information (eg date, cost) as well as the source of energy (solar, wind etc). Green also purchases energy on the wholesale market, ensuring that any energy purchased comes with a renewable energy certificate to ensure it is from a renewable source. Any purchase from the wholesale market automatically gets classed as solar power, regardless of its source.

Staff survey

Annually, a survey is sent round all employees to gain feedback on how to improve processes and to determine overall satisfaction. The survey features a rating system (1 to 5 stars) on a variety of questions and 92% of staff completed the survey this year. All the responses are shown in a password-protected read only .csv file, with each row representing a staff member and their individual responses to the questions included in the survey. Some employees did highlight that they had to click 'submit' a number of times before the survey would go to the completion page. The IT department suspects there was a bug in the system which was causing this issue and are currently investigating the matter.

Question 2

You have recently joined the internal audit department of Spin Home Bikes Ltd (Spin). You previously worked in a large audit firm where you were a data champion due to your experience in using automated tools and techniques, including audit data analytics (ADAs).

Spin is an exercise bike manufacturer based in Manchester, England. The company manufactures exercise bikes which it sells to individuals as well as gyms around the UK. Spin has a large manufacturing plant and a head office on the outskirts of Manchester and employs 340 people. Spin has an excellent reputation in producing high-quality exercise bikes, starting with its original product, Turbo, which can be found in many gyms around the UK.

Financial statement audit

Spin's Finance Director, Ewa Szubert, has advised you that Spin's external auditor is due to commence their planning for the external audit for the year ended 31 March 20X3. As part of the audit work for this year, the external audit team plans to use ADAs to evaluate and test the balances concerning payroll and directors' remuneration. The external audit team is planning to place reliance on Spin's payroll system (SPS), which was designed internally 2 years ago. SPS has the ability to run custom reports on areas such as joiners and leavers and departmental employee numbers. All reports can be run for any selected timeframe, but typically show annual or monthly information.

You have been provided with some outputs from SPS, including screenshots from reports generated, as well as an extract of raw data from the payroll system. Ewa would like you to review this information and highlight any potential concerns over the validity, accuracy and completeness of the data. She would also like you to evaluate the impact that this may have on the external auditor's work. Where you have identified issues, Ewa would also like you to note, where applicable, any further information the external auditor may request, or steps Spin should take to rectify any issues identified ahead of the audit team commencing their fieldwork for the final audit in May 20X3.

Questions:

1. Analyse and evaluate the results of the outputs from SPS to highlight any concerns over the validity, accuracy and completeness of the data relating to payroll and the impact this may have on the external auditor's work; and

2. Describe, where applicable, what further information the external auditor may request or the steps that Spin should take to rectify any issues identified.

(25 marks)

SPS outputs provided by Ewa Szubert

Raw data extract from payroll system for 12 months to 30 November 20X2

Report summary: Annual

	EE number	Gross pay	EE Pension	EE NI & PAYE	Net pay	ER pension
Ted Roy	256	£50,200	£4,016	£12,401	£33,783	-
Blair Daniel	59	-	-	-	-	-
Daniel Blair	60	£26,750	£802	£4,897	£21,051	-
Rachel Gillingham	145	£35,000	£2,625	£7,537	£24,838	-
Zoe Aitken	134	£21,800	£1,090	£3,313	£17,397	-
Eleanor Li	286	£47,800	£1,633	£4,780	£41,387	-

Starters and leavers for October 20X2

Report summary: Monthly

Name	Status	Location	Annual salary	Start date
Craig Brookes	Starter	Manufacturing	£30,000	17/10/2022
Hilary Lee	Starter	Finance	£42,500	06/10/2022
Nick Littlejohn	Leaver	Manufacturing	£28,000	-
Hester Roulston	Leaver	HR	£25,000	-
Mo Salmon	Starter	Executive	£80,000	04/09/2022

A bar chart representing number of employees per department as at 30 November 20X2

